

Appendix A

The Blank Slate app is a software tool that integrates spaced retrieval learning techniques with a machine learning model to deliver personalized memory support. Spaced retrieval refers to the process of repeatedly recalling information at increasingly longer intervals to reinforce long-term memory over time.

In a typical Blank Slate session, users spend a few minutes daily answering questions on topics they want to remember. These questions come in various formats, such as multiple-choice, true/false, and text-entry. After each response, users receive immediate feedback on their accuracy and, if an answer is incorrect, the app displays the correct one. The app requires users to answer all questions correctly before completing each session, after which the user is not prompted to engage with the app until the self-selected time the following day.

The app uses a custom algorithm that predicts the likelihood of a user forgetting each specific question, such that the app prompts users to review only the questions they are most at risk of forgetting. If no significant risk of forgetting is detected on a given day, the app won't prompt any engagement. This ensures a personalized schedule of spaced repetitions tailored to each user's unique forgetting rate. Screenshots of the app are provided below for reference.

